Quantifying linguistic structure's influence on implicit and explicit prosody in developing readers

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Background

- Prior work demonstrates that **prosodic fluency** predicts **reading comprehension** in both primary school readers [1] and high school readers [2] but the causal connection is unclear
- The current ongoing study assesses how **implicit prosody** [3], as measured by silent reading durations, predicts **explicit prosody,** as measured by spoken durations, *of the same text*
- **(1)** If children are realizing **implicit prosody**, we predicted that **psycholinguistic predictors** of spoken durations while **reading aloud,** and first pass fixations while **reading silently,** would correlate
- **(2)** Moreover, we predicted that, across individuals, reading comprehension skill would correlate with the predictions of metric and syntactic structure models when either reading aloud or reading silently

Data Collection

- Participants:
  - Data have been collected from 128 participants of which 124 have been fully processed: 18 are excluded as non-readers; 12 are excluded for incomplete data; **94 are analyzed here**
- Materials:
  - *The Cat in the Hat* [3] has a consistent metric and rhyme structure, simple syntax, & primarily mono-syllabic high frequency words
  - It is comprised of 72 4-line rhyming stanzas
  - The text was presented without pictures (FIGURE 1)

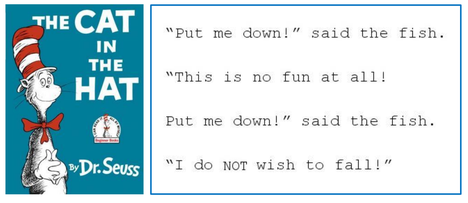


FIGURE 1: *The Cat in the Hat* book cover and one page of text as presented during the experiment

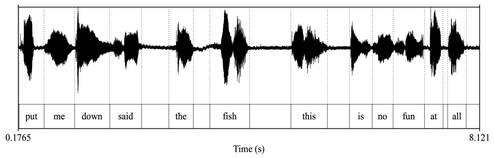
- Procedure:
  - Day 1:
    - Participants read the book aloud into a head-mounted microphone
      - Read the book as if reading to a parent or teacher
      - If they encounter an unknown word, sound out or skip
      - Trials time out after 60 seconds; participants who timed out of 10 or more trials were excluded from analysis as non-readers
  - Day 2: At least one week later:
    - Participants read the book while their eyes are tracked
      - Trials time out after 60 seconds; participants who timed out of 10 or more trials were excluded from analysis as non-readers
    - Participants also complete standardized assessments of reading comprehension (KTEA) [6], phonological awareness and processing speed (CTOPP2) [7]

Text Modeling

	Put	me	down	said	the	fish	this	is	no	fun	at	all	Put	me	down	said	the	fish	I	do	NOT	wish	to	fall
Metric Level	1	1	2	1	1	3	1	1	2	1	1	4	1	1	2	1	1	3	1	1	2	1	1	5
Syntactic Structure	1	3	6	1	1	6	5	1	2	3	1	6	1	3	6	1	1	6	5	1	2	1	1	6
Word Length	3	2	3	3	2	3	3	2	2	3	2	2	3	2	3	3	2	3	1	2	3	3	2	3
Word Frequency	6.1	7.1	6.8	7.6	11.2	3.6	8.5	9.2	7.7	3.8	8.6	8	6.1	7.1	6.8	7.6	11.2	3.6	8.6	7.2	8.4	4.7	10.2	5
Word Class	1	0	0	1	0	1	0	0	0	1	0	0	1	0	0	1	0	1	0	0	0	1	0	1
Font Emphasis	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
Intra-stanza Repetition	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	0	0	0	0	0	0

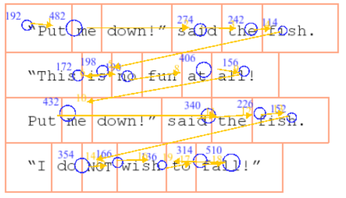
- The text of the book was modeled with a set of linguistic predictors [as in 9]
- Metric and syntactic structure are modeled with hierarchical predictors
- Additional predictors account for lexical and discourse features

Acoustic Analysis



- Each participant's productions were first cleaned of disfluencies, errors, repetitions
- Productions were aligned with the book text using the Montreal Forced Aligner [7]
- Alignments were checked and corrected, & each word's duration was extracted [8]
- Production durations greater than 3 SD's above or below each participants' mean were excluded

Eye-tracking Analysis



- Eye position was sampled at 1000Hz with an EyeLink 1000+
- Each participant's fixations were first cleaned of blinks
- Fixations < 80 and > 1200ms were excluded
- Fixation durations > 3 SD's above or below each participants' mean were excluded

Acknowledgments / References

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Pictures by Leon Nguyen for the Springfield Republican

Results

(1) For each participant, regression models were fit predicting **spoken** and **silent** reading durations from well-attested predictors of reading duration [9]. For each pair of predictors' coefficients in the spoken and first pass models, a Pearson correlation compared the strength of each predictor (TABLE 1).

ACOUSTIC MODEL

Spoken duration ~ Meter Level + Syntactic Level + Number of phonemes + Word Class + Intra-stanza Repetition + Capitalization + Lexical frequency

FIRST PASS MODEL

First pass reading ~ Meter Level + Syntactic Level + Number of phonemes + Word Class + Intra-stanza Repetition + Capitalization + Lexical frequency

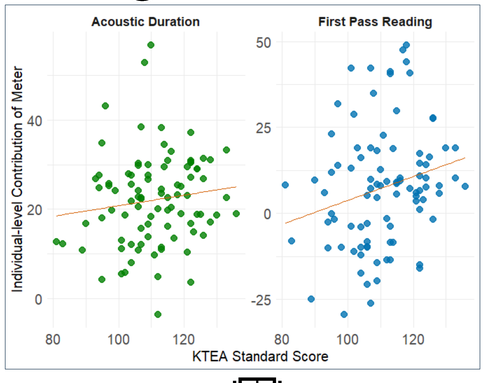
Predictor	Pearson's R	P-value (Holm)
Intra-stanza repetition	0.44	**< .001**
Number of phonemes	0.38	**< .001**
Word class	0.28	**0.02**
Syntactic Level	0.24	**0.06**
Capitalization	0.23	**0.06**
Lexical Frequency	0.12	0.46
Meter Level	-0.01	0.86

TABLE 1: Strength of Pearson correlation between psycholinguistic predictors of first pass and spoken reading durations. P-values are corrected for multiple comparisons.

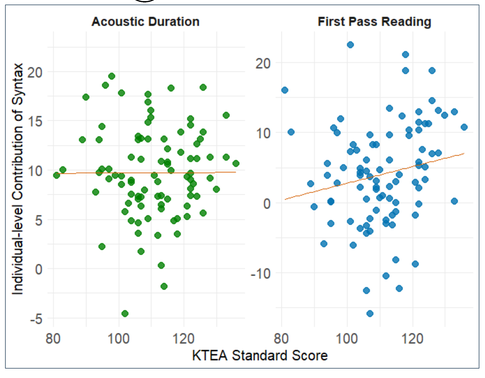
RESULT: The correlations between predictor coefficients for **intra-stanza repetition, word class, and word length were significant**, while the correlations between the coefficients for **syntactic structure and capitalization were marginal**.

(2) We compared the relationship between individual-level coefficients for metric and syntactic structure in the two conditions and their relationship with reading comprehension:

Metric Structure Predicts Reading Skill



Syntactic Structure Predicts Reading Skill



RESULT: Better comprehenders exhibited higher coefficients **for metric structure** in both **spoken durations** and **first pass durations**; coefficients for **syntactic structure** increased with **comprehension** for **first pass fixations**, but not for **spoken durations**.

Conclusions

- **Support for Implicit Prosody**: developing readers who read the same text **aloud** and **silently** **are affected by similar psycholinguistic predictors.** Intra-stanza repetition, length, word class, syntactic structure, and capitalization predictors are strongly correlated
- **Prosodic fluency predicts comprehension:**
  - Better comprehenders' reading durations reflect hierarchical metric structure in both reading contexts
  - Better comprehenders' reading durations better reflect hierarchical syntactic structure only when reading silently